

JEE MAIN REVISION MASTER: AREA UNDER THE CURVES 2026 1 QUESTIONS 1.67% WEIGHTAGE

↓49.85%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 17 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Area Under The Curves — Standard Calculations & Problem Archetypes	9 PYQs	Quantitative Analysis
Moderate	Area Under The Curves — Core Principles & Governing Laws	4 PYQs	Core Principle
Moderate	Area Under The Curves — Applied Cases, Reactions & Traps	4 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Area Under The Curves — Standard Calculations & Problem Archetypes	$y = -\sqrt{16 - x^2}$, $\quad y = x^2$, $\quad \int_0^{\pi/4} (\cos x - \sin x) dx$
Area Under The Curves — Core Principles & Governing Laws	$y = x$, $\quad y = x^2$, $\quad y = 16$
Area Under The Curves — Applied Cases, Reactions & Traps	$x = 0$, $\quad y = 2x$, $\quad y^2 = 8x$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Area Under The Curves — Standard Calculations & Problem Archetypes (High Yield — 9 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Area Under The Curves.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The area of the region bounded by the curve $y^2=x^3$, the y -axis and the lines $y=1$ and $y=8$ is..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "The area of the region bounded by the line $y=x$ and the curve $y=x^3$ is..." [Ans: N/A]
- [2024 Shift] Q#4 (MCQ): "The area of the region bounded by the line $y=3x$ and the curve $y=x^2$ sq units is..." [Ans: N/A]

3.2 Area Under The Curves — Core Principles & Governing Laws (Moderate Yield — 4 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Area Under The Curves.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Area Under The Curves.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Area Under The Curves PYQs.

Representative Source PYQs:

- [2025 Shift] Q#3 (MCQ): "The area of the region bounded by the curve $y=x^2$ and the line $y=16$ is..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "The area of the region bounded by the line $y=x+1$ and the lines $x=3$ and $x=5$ is..." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "The area of the region bounded by the line $y=2x+1$, X -axis and the ordinates $x=-1$ and $x=1$ is..." [Ans: N/A]

3.3 Area Under The Curves — Applied Cases, Reactions & Traps (Moderate Yield — 4 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Area Under The Curves.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2025 Shift] Q#2 (MCQ): "The area bounded by the curve $y=\sin((x/3))$, x axis, the lines $x=0$ and $x=3\pi$ is..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "In the interval $(0,\pi/2)$ area lying between the curves $y=\tan x$ and $y=\cot x$ and the X -axis is..." [Ans: N/A]
- [2020 Shift] Q#2 (MCQ): "The area of the region bounded by the curve $y^2=8x$ and the line $y=2x$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The area of the region bounded by the curve $y^2 = x^3$, the y-axis and the lines $y=1$ and $y=8$ is

- (A) $(155 / 3)$ sq. units
- (B) $(93 / 5)$ sq. units
- (C) 93 sq. units
- (D) 155 sq. units

Correct Answer: None

Solution Analysis: The curve is $y^2 = x^3$. Since the region is between the lines $y=1$ and $y=8$, it is easier to write x in terms of y . From $y^2 = x^3$ we get $x = y^{2/3}$. The y-axis is $x=0$. So, for each value of y from 1 to 8, the horizontal width of the region is $x_{\text{right}} - x_{\text{left}} = y^{2/3} - 0 = y^{2/3}$. Hence, the required area is $\int_1^8 (1 - 0) y^{2/3} dy$. Now, $\int y^{2/3} dy = \frac{3}{5} y^{5/3} = \frac{3}{5} y^{5/3}$. So, Area = $\left[\frac{3}{5} y^{5/3} \right]_1^8 = \frac{3}{5} (8^{5/3} - 1^{5/3})$. Since $8^{1/3} = 2$ we have $8^{5/3} = 2^5 = 32$. Thus, Area = $\frac{3}{5} (32 - 1)$.

Benchmark Problem 2: KCET 2025 (Q#2)

The area bounded by the curve $y = \sin((x / 3))$, x-axis, the lines $x=0$ and $x=3\pi$ is

- (A) 9 sq. units
- (B) $(1 / 3)$ sq. units
- (C) 6 sq. units
- (D) 3 sq. units

Correct Answer: None

Solution Analysis: The required area is found by integrating the function $y = \sin((x / 3))$ from $x=0$ to $x=3\pi$. This gives the definite integral: $\int_0^{3\pi} \sin((x / 3)) dx$. To integrate $\sin((x / 3))$, we use substitution. The integral becomes $-3 \cos((x / 3))$. Now substitute the limits 0 and 3π : $\int_0^{3\pi} \sin((x / 3)) dx = (-3 \cdot \cos(x / 3)) \Big|_0^{3\pi} = (-3(-1)) - (-3(1)) = 6$. So, the total area is 6.

Benchmark Problem 3: KCET 2025 (Q#3)

The area of the region bounded by the curve $y = x^2$ and the line $y=16$ is

- (A) $(32 / 3)$ sq. units
- (B) $(256 / 3)$ sq. units
- (C) $(64 / 3)$ sq. units
- (D) $(128 / 3)$ sq. units

Correct Answer: None

Solution Analysis: $y = x^2, y = 16$. $2 \int_0^{16} \sqrt{y} dy = 2 \left[\frac{2}{3} y^{3/2} \right]_0^{16} = \frac{4}{3} (4^3) = \frac{256}{3}$